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Service Accessibility Barriers and Support Priorities Reported by Caregivers of Children with Neurodevelopmental Disorders in Jazan, Saudi Arabia

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ABSTRACT

Evidence on caregiver-reported service accessibility, support needs, and priorities for children with neurodevelopmental disorders in Jazan remains limited. The main objectives of this study are to quantify caregiver-reported access barriers, describe informational and social support indicators and service priorities, and identify factors associated with access problems. We conducted a cross-sectional survey of caregivers in Jazan between February and July 2025 (n = 217). Access outcomes in the past 12 months were local unavailability, waiting lists/appointment delays, financial difficulty, and difficulty obtaining information. The survey also captured caregiver-reported informational and social support indicators (e.g., awareness/participation in advocacy groups, information sources, and satisfaction with supports) and priorities for service improvement. Associations were examined using bivariate tests (two-sided $P < 0.05$). Children had a mean age of 7.0 years; first concerns arose at 2.7 years and diagnosis occurred at 4.5 years. Hospital-recorded neurodevelopmental disorder diagnoses included attention-deficit/hyperactivity disorder (ADHD; 72.4%), developmental speech and language disorder (27.2%), intellectual disability (25.8%), and autism spectrum disorder (13.4%). Caregivers reported waiting lists (53.5%), local unavailability (50.2%), financial difficulty (48.4%), and difficulty obtaining information (47.0%). Advocacy-group awareness was low (11.5%), with 6.5% reporting participation. Later diagnosis was associated with local unavailability (P = 0.036) and waiting list delays (P = 0.019). Intellectual disability was associated with financial difficulty (P = 0.037) and difficulty obtaining information (P = 0.009); ADHD was associated with financial difficulty ($P < 0.001$) and difficulty obtaining information (P = 0.001). Caregivers most frequently prioritized improved education services (90.8%), followed by healthcare (55.3%), parent education at diagnosis (46.1%), and community awareness (39.2%). In conclusion, access barriers are common and linked to delayed diagnosis, while informational and social supports remain limited. Findings support coordinated cross-sector action to expand local capacity and strengthen family-centered supports.

KEYWORDS

Neurodevelopmental disorders, Health services accessibility, Caregiver needs, Child, Saudi Arabia

INTRODUCTION

Neurodevelopmental disorders (NDDs) are conditions that begin early in life and affect cognition, communication, behavior, motor skills, and social interactions (Smythe *et al.,* 2024). They are formally defined as disorders with onset during the developmental period and characterized by limitations in adaptive functioning across academic, occupational, and social domains (American Psychiatric Association, 2022). The International Classification of Diseases, 11th Revision (ICD-11), includes disorders of intellectual development, autism spectrum disorder (ASD), developmental speech and language disorders, developmental learning disorders, developmental coordination disorder, and attention-deficit/hyperactivity disorder (ADHD), among others (World Health Organization, 2021). These conditions often require sustained health, educational, and social support, placing significant demands on families and caregivers (Smythe *et al.,* 2024).

Globally and regionally, NDDs contribute substantially to disability and to health, education, and social service demand. Global reporting often uses the umbrella term

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"developmental disabilities" to describe a broad group of developmental conditions that include several NDDs. Using this framing, the Global Burden of Disease (GBD)–World Health Organization (WHO) Rehabilitation Database. Rehabilitation Database estimates that over 53 million children under 5 years of age live with developmental disabilities, with the highest burden in low- and middle-income countries where early detection and services are limited (Olusanya *et al.,* 2022; World Health Organization and UNICEF, 2023). In the United States, national health survey data indicate that 17.3% of children aged 3-17 years have at least one developmental disability and 6.7% have two or more, and these children have higher health and therapy service needs than peers (Cogswell *et al.,* 2022). Within NDDs, ADHD is common globally, with meta-analytic estimates indicating a prevalence of 7.6% in children and 5.6% in adolescents (Salari *et al.,* 2023). In the Arab region, ADHD prevalence is similarly high in pooled estimates, including 10.3% overall in a Middle East and North Africa meta-analysis and 12.4% across 14 studies in a Saudi Arabia-specific meta-analysis (Aljadani *et al.,* 2023; Al-Wardat *et al.,* 2024). For ASD, pooled estimates are generally lower but vary by methodology, with higher estimates reported in studies conducted after 2015 and marked differences in Saudi prevalence estimates depending on whether registry-based data or community screening was used (Sabbagh *et al.,* 2021; AlBatti *et al.,* 2022; Akomolafe *et al.,* 2025). In Jazan, hospital-based data also point to substantial clinical demand: ADHD (31.5%), ASD (27.6%), and intellectual disability (24%) were among the most common diagnoses in children and adolescents receiving care (Meraya, 2020).

In Jazan, we examine caregiver-reported service accessibility through four access barriers: local service unavailability, waiting list or appointment delays, financial difficulty, and difficulty obtaining information, alongside caregiver-reported support indicators (information sources, advocacy-group awareness or participation, and satisfaction with supports). In Saudi Arabia, reviews highlight persistent barriers affecting families' access to specialized health and education supports, including limited-service availability, workforce constraints, and gaps in coordination that hinder inclusive provision (Alanazi, 2020; Zaino, 2025). Despite growth in disability-related publications, bibliometric evidence identifies gaps in Saudi service delivery research relevant to improving access and family support (Kandeel *et al.,* 2025). Caregivers of children with NDDs often report psychological stress, financial strain, and reduced quality of life (Vilaseca *et al.,* 2023; Vukićević *et al.,* 2023). These challenges may be amplified by contextual constraints in Jazan, including a substantial rural population (approximately 40%) and centralized specialist service provision, which can make specialized services harder to reach and navigate (Meraya, 2020).

Evidence on caregiver-reported access barriers and support gaps for children with NDDs in Jazan remains limited. This study therefore quantifies caregiver-reported barriers, examines bivariate factors associated with access problems, and describes support indicators and caregiver priorities. The focus is service accessibility and caregiver-identified needs, not prevalence estimation, and the results are intended to inform regional planning and Vision 2030-aligned reforms for more inclusive health and education services.

METHODS

Study design and setting

This cross-sectional survey was conducted in the Jazan region of Saudi Arabia between February and July 2025 at the province's main referral center for child and adolescent psychiatry, which provides specialized outpatient mental health services in the region (Meraya, 2020). The study described caregiver-reported service accessibility, support indicators, and priorities for service improvement in a setting where specialist care is centralized and alternative service pathways are limited.

Participants and recruitment

Participants were parents or primary caregivers of children and adolescents with a hospital-recorded NDD diagnosis who attended the study center during the data collection period. Caregivers were approached consecutively at the end of the clinical encounter by trained physician data collectors. Participation was voluntary, and eligible caregivers were included after providing written informed consent and completing the questionnaire.

Sample size

A pragmatic census approach was used, inviting all eligible caregivers during the study window; therefore, no a priori power calculation was undertaken, which we acknowledge as a limitation. To aid interpretability, we report indicative detectable effect sizes rather than using power to justify sample adequacy. With $n = 217$ and $\alpha = 0.05$, detectable effects are approximately Cohen's $d \approx 0.40$ for balanced two-group mean comparisons and Cramér's $V \approx 0.24$ for 2×2 contingency tables; power is lower for unbalanced groups or sparse tables. The primary aim was descriptive estimation.

Instrument development and measures

A structured questionnaire was adapted from two sources: the Caregiver Needs Survey used in the Erasmus+ SynergyEd project (Croatia, North Macedonia, Serbia) and the U.S. National Survey of Children with Special Health Care Needs (NS-CSHCN 2009-2010). Items were integrated to capture caregiver experiences with services, perceived support, barriers, unmet needs, and family priorities. The research team reviewed items for cultural relevance while preserving the original constructs.

The English version was translated into Arabic using forward and backward translation by two bilingual pharmacists, with adjudication by a third reviewer. A brief cognitive pilot

with caregivers (n = 10) confirmed clarity and cultural relevance, and minor wording edits were made before final implementation. Formal psychometric revalidation was not undertaken, as the instruments were adapted for descriptive use and not for scale-based inference.

Service accessibility (primary outcomes)

Four caregiver-reported outcomes were analyzed (each coded yes/no): local service unavailability, waiting list or appointment delays, unaffordable costs, and difficulty obtaining adequate information.

Support indicators, caregiver impact, and priorities

The questionnaire also assessed information sources, support indicators (including advocacy awareness/participation), caregiver burden and family impact, satisfaction with supports, and caregiver-identified priorities for service improvement.

Hospital-recorded diagnoses and derived variables

Hospital-recorded diagnoses were classified as NDDs based on ICD-11 categories, including disorders of intellectual development, developmental speech and language disorder, ASD, ADHD, developmental learning disorder, developmental motor coordination disorder, and other specified or unspecified NDDs. Diagnoses were not mutually exclusive; children could have more than one NDD recorded. A derived variable captured diagnostic complexity, categorized as a single diagnosis (one category) versus multiple diagnoses (two or more categories).

Covariates and care arrangements

Covariates included caregiver, child, and household characteristics and care arrangements (none, medication only, therapy only, or combined medication and therapy). Full item content and coding are provided in Supplementary Appendix A (protocol v1.0).

Data collection procedures

Physician data collectors received standardized training on recruitment, questionnaire administration, and ethical procedures. Caregivers were invited to participate after the clinical encounter; written informed consent was obtained before questionnaire completion (median completion time: approximately 30 min). Questionnaires were completed on paper, stored securely on-site, and then entered into a secure electronic data capture form for data management (see Supplementary Appendix B for the detailed data collection workflow/protocol).

Ethical considerations

Ethical approval was obtained from the Jazan Health Cluster Ethics Committee, Saudi Arabia (approval no. 24117; December 25, 2024; National Registration Number H-10-Z-141). All procedures followed the Declaration of Helsinki (2013 revision). Caregivers received study information, provided written informed consent, and were informed that participation was voluntary with the right to withdraw at any time. No personally identifiable information was collected; questionnaires were stored securely and data were de-identified prior to analysis.

Statistical analysis

Analyses were conducted in Stata 18.0 (StataCorp, College Station, TX, USA; RRID:SCR_012763); no custom code was written. Categorical variables are summarized as counts and percentages, and continuous variables as mean [standard deviation (SD)]. All tests were two-tailed with $\alpha = 0.05$. For transparency, test statistics, degrees of freedom, and exact P-values are reported in Supplementary Tables S1-S4.

The four service accessibility outcomes (yes/no) were analyzed as binary dependent variables. Bivariate analyses explored associations with caregiver- and child-level characteristics. Multivariable analyses were not undertaken because the study aim was descriptive and exploratory, and some diagnostic subgroups had small cell sizes, which would have limited model stability and interpretability. Continuous variables included caregiver age, child age, age at first concerns, and age at diagnosis. All other variables, including binary and ordinal survey responses, were analyzed as categorical.

Missing data were handled using complete-case analysis at the comparison level; no imputation was performed, and item-specific denominators are reported in Tables 1–4. This approach was used because the study aim was descriptive and missingness occurred as item-level nonresponse; however, complete-case analysis may introduce bias if missingness is related to outcomes or covariates.

Pearson's chi-square test was used for categorical associations (Fisher's exact test when expected cell counts were small, e.g., <5). Categories were collapsed when needed to address sparse cells and improve interpretability. Independent-samples t-tests were used for mean comparisons. Formal assumption testing was not conducted routinely; therefore, inferential results are interpreted cautiously and presented as exploratory.

To contextualize the magnitude of effects that could be detectable in this sample, we provide indicative effect-size benchmarks as sample size-based reference values. We report P-values for the observed associations; however, Cohen's d (for two-group mean comparisons) and Cramér's V (for 2 × 2 categorical associations) were not calculated or reported for individual study associations. Because a pragmatic census approach was used, we did not perform an a priori power calculation. Accordingly, the benchmark values presented reflect only the achieved

sample size and are intended solely to support interpretability in this primarily descriptive study, rather than to imply confirmatory hypothesis testing or to provide effect-size estimates for specific comparisons.

RESULTS

Sample characteristics

A total of 217 caregivers participated. The mean caregiver age was 40.1 (SD 8.9) years. Most caregivers were female (77.4%) and married (89.9%). Reported parental consanguinity was common, with 32.3% reporting first-cousin marriage. Caregiver education was primary (15.7%), secondary (44.7%), bachelor's (37.8%), and postgraduate (1.8%). Monthly household income was most frequently 5001-10,000 Saudi riyals (SAR; 38.7%) or 10,001-15,000 SAR (32.3%). Full sample characteristics are summarized in Table 1.

Table 1: Caregiver demographic characteristics.

Variable	Summary
n	217
Age, years [mean (SD)]	40.11 (8.88)
Sex	
Male	49 (22.6%)
Female	168 (77.4%)
Education	
Primary school	34 (15.7%)
Secondary school	97 (44.7%)
Bachelors	82 (37.8%)
Masters/PhD	4 (1.8%)
How many children do you have? [mean (SD)]	3.96 (2.79)
How many children with neurodevelopmental disorders do you have? [mean (SD)]	1.11 (0.38)
Marital status	
Married	195 (89.9%)
Separated/divorced	20 (9.2%)
Widowed	2 (0.9%)
Spouse education	
Primary school	45 (20.9%)
Secondary school	92 (42.8%)
Bachelors	75 (34.9%)
Masters/PhD	3 (1.4%)
Spouse relative	
First cousins	70 (32.3%)
Second cousins	15 (6.9%)
Distant cousins (beyond second)	12 (5.5%)
Not related by blood	120 (55.3%)
Family income	
<5000 SAR	34 (15.7%)
5001-10,000 SAR	84 (38.7%)
10,001-15,000 SAR	70 (32.3%)
15,001-20,000 SAR	22 (10.1%)
>20,000 SAR	7 (3.2%)

Abbreviations: SAR, Saudi riyals; SD, standard deviation.

Child concerns, diagnoses, and care received

Children had a mean age of 6.95 years. Caregivers first noticed developmental concerns at a mean age of 2.67 years, and a formal diagnosis was made at 4.53 years, reflecting an average interval of 1.9 years between first concern and diagnosis.

Caregivers most frequently reported early motor or coordination concerns (63.1%), late talking (48.4%), and not responding to sounds or name (45.6%). Patterns of early concerns are summarized in Table 2.

Diagnoses were not mutually exclusive, and children could have more than one hospital-recorded NDD; therefore, percentages may exceed 100%. The most common hospital-recorded diagnoses were ADHD (72.4%), developmental speech and language disorder (27.2%), and disorders of intellectual development (25.8%); ASD accounted for 13.4% (Table 2). The most frequent diagnostic combinations were developmental speech and language disorder plus ADHD (n = 25) and disorders of intellectual development plus ADHD (n = 16).

Regarding care, 44.7% received medication plus occupational therapy and 29.0% received medication only. Most children were diagnosed by a psychiatrist (74.2%) and received care through a government hospital (88.0%). Educational placement most commonly included special schools (30.0%), public schools (24.0%), and preschools (24.0%); 18.0% were not enrolled. Additional academic support was reported by 31.8%, and 40.7% reported special assistance from a government organization (Table 2).

Social support and information sources

Caregivers reported multiple sources of information, including healthcare professionals (including physicians and other providers) and online resources (Table 3). Specifically, information sources most commonly included other providers such as health specialists and therapists (53.9%), the internet (50.7%), and child's physician (30.0%), while teachers (20.7%) and other parents of children with similar conditions (10.1%) were less frequently reported. Awareness of advocacy groups was low (11.5% aware; 6.5% participating). Other sources included informal community contacts and non-specific online platforms. Levels of satisfaction with available information and perceived adequacy of support varied across respondents. These information-seeking patterns were examined further in relation to reported barriers to service access.

Service access and unmet needs

Caregivers reported substantial barriers to accessing services, including waiting lists or appointment delays (53.5%), local service unavailability (50.2%), financial difficulty (48.4%), and difficulty obtaining adequate information (47.0%) (Table 4). Barriers were not mutually exclusive, indicating that many families experienced

Table 2: Child's demographic statistics.

Variable	Summary
n	217
Age, years [mean (SD)]	6.95 (3.01)
Age at concern, years	2.67 (1.34)
Age at diagnosis, years	4.53 (2.38)
Sex	
Male	162 (74.7%)
Female	55 (25.3%)
Behavior reasons of concerns	
Medical issues (e.g., seizures, poor growth, recurrent stomach problems)	28 (12.9%)
Limited eye contact during conversation or play	50 (23.0%)
No response to name or to sounds	99 (45.6%)
Difficulty reading/using non-verbal cues (tone of voice, facial expressions, body language)	47 (21.7%)
Coordination/gross motor difficulties (e.g., walking problems)	137 (63.1%)
Speech started later than typical for age	105 (48.4%)
No spoken words at the time	23 (10.6%)
Speech less advanced than same-age peers	44 (20.3%)
Loss of previously acquired speech skills	60 (27.6%)
Did not seem to understand spoken instructions	82 (37.8%)
Fine motor difficulties (e.g., using scissors, drawing with crayons)	23 (10.6%)
Challenges in playing or interacting with other children	21 (9.7%)
Difficulty acquiring self-care skills (e.g., toilet training, dressing)	41 (18.9%)
Difficulty learning early academic concepts (e.g., alphabet, numbers)	62 (28.6%)
Child's clinical diagnosis in the hospital records	
Disorders of intellectual development	56 (25.8%)
Developmental speech and language disorders	59 (27.2%)
Autism spectrum disorder	29 (13.4%)
Developmental learning disorder	12 (5.5%)
Developmental motor coordination disorder	1 (0.5%)
Attention-deficit/hyperactivity disorder	157 (72.4%)
Other specified neurodevelopmental disorder	1 (0.5%)
Other	12 (5.5%)
Therapy	
None	25 (11.5%)
Medications only	63 (29.0%)
Occupational therapy only	32 (14.7%)
Medications + occupational therapy	97 (44.7%)
Other form of therapy	
No	192 (88.5%)
Yes	25 (11.5%)
Who diagnosed the child?	
Primary care physician/family physician	11 (5.1%)
Pediatrician	25 (11.5%)
Pediatric specialist (i.e., developmental pediatrician)	7 (3.2%)
Psychiatrist	161 (74.2%)
Neurologist	6 (2.8%)
Team of professionals	3 (1.4%)
Other	4 (1.8%)
Source of care	
Government hospital	191 (88.0%)
Specialist centers for children with special needs	24 (11.1%)
Both	2 (0.9%)

Table 2: Continued.

Variable	Summary
What type of school is your child attending at present?	
Preschool	52 (24.0%)
Government/public school	52 (24.0%)
Private school	9 (4.1%)
Special education school (for children with disabilities)	65 (30.0%)
Not currently enrolled/none of the above	39 (18.0%)
Is your child receiving extra academic support related to his/her developmental needs?	
No	148 (68.2%)
Yes	69 (31.8%)
Is your child currently receiving any special assistance related to a developmental condition?	
None	128 (59.3%)
Government organization	88 (40.7%)

Abbreviation: SD, standard deviation.

multiple simultaneous constraints. Despite these reported barriers, recent healthcare utilization was low across several service categories: most families reported no emergency department visits (81.6%), no routine check-ups (76.0%), and no preventive dental visits (79.3%) in the past 12 months (Table 4). Despite high reported need and substantial access barriers, recent healthcare utilization was low, suggesting potential unmet need rather than lack of demand.

Factors associated with service barriers

Bivariate analyses identified several caregiver and child characteristics associated with specific access barriers (Supplementary Tables S1-S4). Formal assumption tests for independent-samples *t*-tests and chi-square tests were not routinely performed; therefore, these bivariate findings should be interpreted cautiously and viewed as exploratory. Effect sizes were not calculated for each bivariate association; instead, indicative effect-size benchmarks based on the achieved sample size are provided in the Sample Size section to aid interpretation.

Local service unavailability was reported more often among caregivers whose children had a later mean age at diagnosis ($P = 0.036$). Caregivers reporting service unavailability were also more likely to report advocacy participation ($P = 0.006$) and to use other parents as an information source ($P = 0.026$) (Supplementary Table S1).

Waiting lists/appointment delays were more common among families with a later mean age at diagnosis ($P = 0.019$) and among caregivers who were aware of advocacy groups ($P = 0.016$). These caregivers more frequently reported relying on other providers ($P = 0.010$) and other non-specific sources ($P < 0.001$) for information (Supplementary Table S2).

Financial difficulty accessing services was more frequently reported when spouse education was primary ($P = 0.044$), and it varied by diagnosis and care characteristics. Financial difficulty was more frequently reported when disorders of intellectual development were recorded

Table 3: Social support and sources of information.

Variable	Summary, *n* (%)
n	217
Are you aware of any advocacy group for children with neurodevelopmental disorders?	
No	192 (88.5%)
Yes	25 (11.5%)
Are you or any household member currently involved with a family support or advocacy group related to your child's developmental condition?	
No	203 (93.5%)
Yes	14 (6.5%)
Source of information	
Internet	110 (50.7%)
Child's physician	65 (30.0%)
Teacher	45 (20.7%)
Other parents of children with neurodevelopmental disorders	22 (10.1%)
Other providers (i.e., health specialists, therapists) who work with my child	117 (53.9%)
Other	131 (60.4%)

(P = 0.037) and when ADHD was recorded (P < 0.001). It also differed by care arrangement, with higher reports among those receiving medication plus therapy compared with therapy only/none (P = 0.010). Caregivers reporting financial difficulty were also more likely to use other parents as an information source (P = 0.016) (Supplementary Table S3).

Difficulty obtaining adequate information was more frequently reported when spouse education was primary (P = 0.022) and among children with disorders of intellectual development (P = 0.009). Difficulty obtaining information was also more common when ADHD was recorded (P = 0.0013). Caregivers reporting difficulty obtaining information were more likely to seek information from other parents (P = 0.003) (Supplementary Table S4). Overall, later age at diagnosis and specific diagnostic profiles were consistently associated with multiple access barriers.

Caregiver burden and family impact

Caregiver frustration with obtaining services was common. Overall, 83.9% reported frustration at least sometimes, including sometimes (47.0%), usually (17.1%), or always (19.8%); 16.1% reported never feeling frustrated. Financial problems related to the child's needs were reported by 43.8% of families. Employment impact was also substantial: 22.6% of caregivers stopped working and 29.5% reduced working hours. For day-to-day caregiving support, 15.2% relied on grandparents and 14.3% on other relatives, while 65.0% reported no additional caregiver beyond the parents (Table 4). These levels of burden likely reflect the cumulative impact of service access barriers, diagnostic delays, and limited support availability.

Satisfaction, challenges, and priorities

Caregiver satisfaction was highest for relationships with service providers (85.7%), support to make progress at home (80.6%), and support from teachers or school-based experts (78.8%). Satisfaction with caregivers' own support from friends was lower (69.1%) (Fig. 1).

Caregivers most frequently identified behavioral difficulties (59.4%), safety concerns (56.2%), and difficulties with daily living (41.0%) as child-level challenges (Fig. 2). At the family level, the dominant challenges were ensuring adequate education (88.9%), healthcare access (74.7%), and welfare or social supports (59.9%) (Fig. 3). For provincial priorities, caregivers most often endorsed improved education services (90.8%), followed by healthcare services (55.3%), parent education at diagnosis (46.1%), and community awareness (39.2%) (Fig. 4).

DISCUSSION

This study provides one of the first systematic, caregiver-centered accounts of service access barriers and support gaps for children and adolescents with NDDs in the Jazan region of Saudi Arabia. Caregiver-reported data were complemented by hospital-recorded diagnoses (based on ICD-11), allowing for cross-validation of reported needs and highlighting associations with specific NDD profiles. By combining hospital-recorded diagnoses with caregiver perspectives, the study highlights key barriers across the care pathway. Families reported service unavailability, long waiting times, financial strain, and difficulty accessing reliable information.

Associations reported in this study are based on bivariate analyses and should be interpreted as exploratory; causal inference is not possible in this cross-sectional design. In a regional context where specialist child and adolescent mental healthcare is largely delivered through a single psychiatric facility (Meraya, 2020), these barriers can place a substantial coordination burden on families as they navigate health and education pathways (Kandeel *et al.,* 2025). By focusing on caregiver-reported accessibility and support needs rather than prevalence estimation, the findings offer actionable evidence for strengthening local capacity and coordination in Jazan and support Saudi Arabia's Vision 2030 priorities for inclusive and equitable health and education services.

Healthcare access difficulties

Barriers to healthcare access were among the most consistent caregiver-reported themes in this study. Many families described services as locally unavailable or difficult to obtain because of waiting lists and appointment constraints, and caregivers often reported relying on alternative providers or informal networks when timely appointments and clear guidance were hard to secure. Half of families (50.2%) reported local service unavailability, and over half (53.5%) experienced appointment delays. Given the centralized

Table 4: Caregiver-reported service access, impact on family, and quality of life.

Variable	Summary, *n* (%)
Access to services	
In the past 12 months, did you experience any delays or difficulty obtaining services for your child due to any of the following?	
Availability (services not available locally)	
No	108 (49.8%)
Yes	109 (50.2%)
Scheduling delays (waiting lists/limited slots)	
No	101 (46.5%)
Yes	116 (53.5%)
Cost (unable to pay/out of pocket)	
No	112 (51.6%)
Yes	105 (48.4%)
Information (hard to find reliable information)	
No	115 (53.0%)
Yes	102 (47.0%)
Other reason	
No	173 (79.7%)
Yes	44 (20.3%)
In the past 12 months, how many times did the child…?	
Visit a hospital emergency room?	
None	177 (81.6%)
1	19 (8.8%)
2-3	13 (6.0%)
≥4	8 (3.7%)
Receive a routine health check-up?	
None	165 (76.0%)
1	39 (18.0%)
2-3	12 (5.5%)
≥4	1 (0.5%)
Receive preventive dental care?	
None	172 (79.3%)
1	25 (11.5%)
2-3	17 (7.8%)
≥4	3 (1.4%)
Impact on caregivers and families	
In the last 12 months, how often did you feel frustrated when trying to obtain services for your child?	
Never	35 (16.1%)
Sometimes	102 (47.0%)
Usually	37 (17.1%)
Always	43 (19.8%)
Has your child's condition resulted in financial difficulties for your family?	
No	122 (56.2%)
Yes	95 (43.8%)
Has anyone in your family had to stop working because of your child's condition?	
No	168 (77.4%)
Yes	49 (22.6%)
Have you or a family member ever reduced work hours owing to your child's condition?	
No	153 (70.5%)
Yes	64 (29.5%)
In addition to the mother/father, who takes care of your child at home?	
Grandparents	33 (15.2%)
Relatives	31 (14.3%)
Neighbors	1 (0.5%)
Maid	19 (8.8%)
None	141 (65.0%)

configuration of specialist child and adolescent mental health services in Jazan (Meraya, 2020), these experiences are consistent with Saudi reports describing long waits and stressful care encounters for families seeking neurodevelopmental services (Al-Jabri *et al.,* 2023; Al Awaji *et al.,* 2024). These barriers illustrate systemic gaps in service decentralization and workforce capacity.

In exploratory bivariate analyses, age at diagnosis showed barrier-specific patterns: later diagnosis was associated with reported service unavailability and waiting list barriers; however, no association was observed for several other access outcomes. These patterns suggest that systemic centralization and diagnostic delays contribute to inequitable access, particularly for children with more complex needs. Families experiencing appointment delays often sought information from alternative providers or informal networks, suggesting compensatory strategies when formal pathways were slow. Notably, greater engagement with advocacy groups among families reporting service unavailability may reflect mobilization when services are perceived as absent rather than merely delayed. International frameworks and regional assessments similarly emphasize clearer pathways and better-coordinated services, reinforcing the policy relevance of strengthening local capacity and navigation support in Jazan (World Health Organization and UNICEF, 2023; Zaino, 2025).

Education access and supports

Educational support gaps were a prominent caregiver-reported concern in this study. Although children were enrolled across public schools, preschools, and special education settings, many caregivers reported that their child was not receiving additional academic support (68.2%), and a minority (18.0%) were not currently enrolled. Education was also the top provincial priority for improvement, underscoring that families view school-based support as central to their child's progress and daily functioning.

Saudi evidence helps explain why these gaps persist, particularly barriers to inclusive education such as limited teacher preparation and resource constraints (Alshamri, 2019; Alanazi, 2020). Caregiver studies in Saudi Arabia likewise emphasize the need for accessible schools, academic modifications, and multidisciplinary, school-linked services that connect special education with therapy supports (Al Awaji *et al.,* 2024), consistent with a broader emphasis on cross-sector collaboration to strengthen support pathways, including education-linked services (Zaino, 2025). These findings support investment in teacher training, inclusive curricula, and coordinated health–education pathways so that educational supports are timely, practical, and aligned with children's functional needs.

Financial and informational barriers

Financial and informational barriers often co-occurred in this sample and were described as practical obstacles to staying engaged in care. Nearly half of caregivers (48.4%)

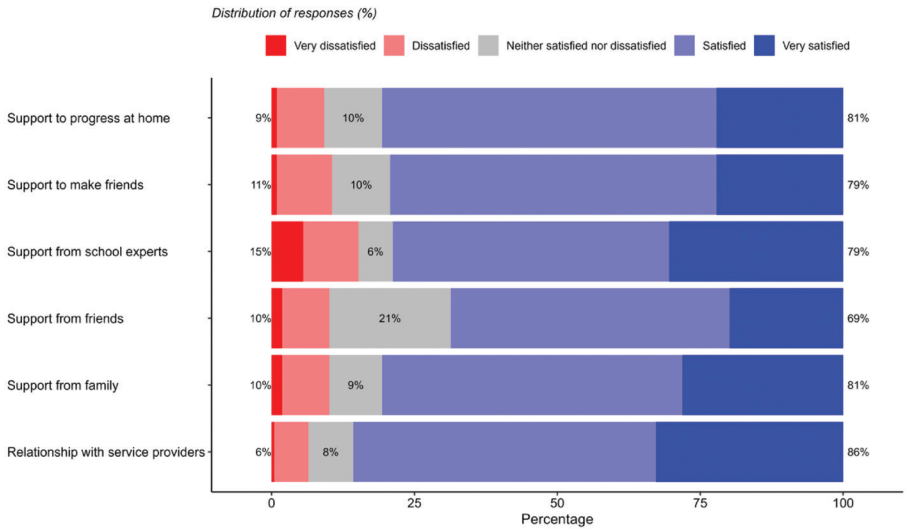

Figure 1: Caregiver-reported satisfaction with selected supports and services in the past 12 months. Stacked 100% bars show the 5-point distribution (very dissatisfied to very satisfied) for six domains: support from teachers/school experts to help the child progress at school, support for progress at home, support for making friends, relationship with service providers who work with the child, support from friends, and support from family. Percentages are based on non-missing responses (overall n = 217; item-level n varies). Values at the right margin indicate the combined satisfied/very satisfied proportion.

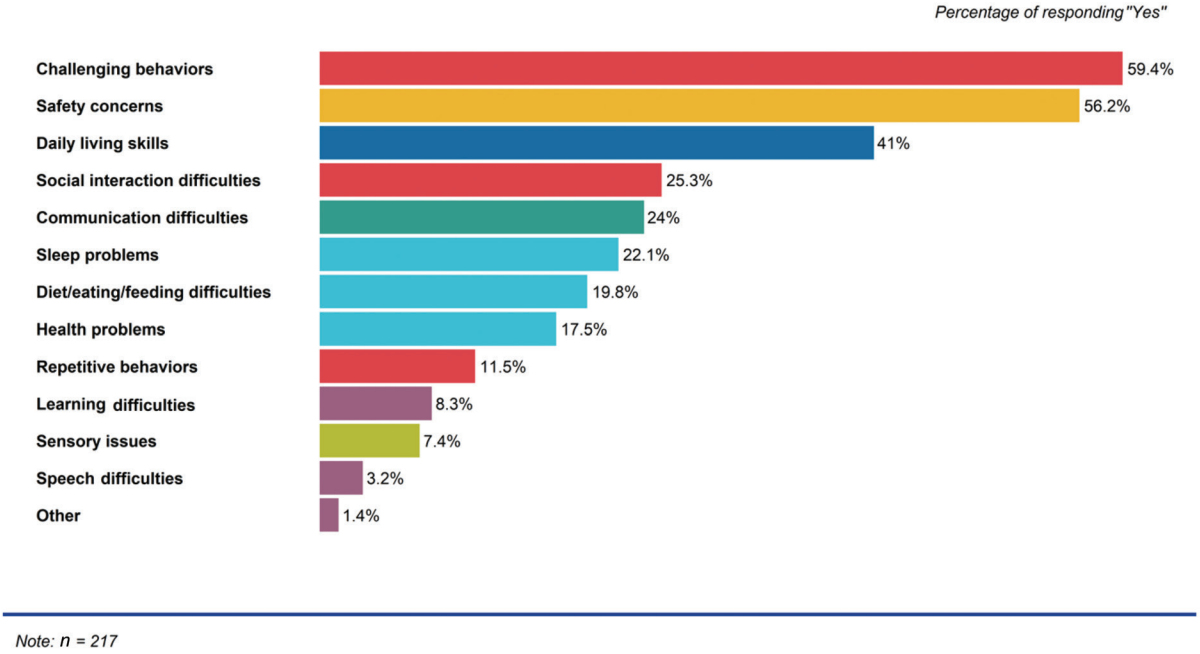

Figure 2: Distribution of caregiver-reported care difficulties for children with neurodevelopmental disorders. Totals exceed 100% because respondents could select any number of options.

reported financial difficulty, with higher prevalence among children receiving combined medication and therapy (P = 0.010). Financial strain was also more commonly reported among families of children with ADHD or disorders of intellectual development and among families where the spouse's education was primary, indicating that both clinical complexity and socioeconomic position may shape affordability. Informational barriers were similarly common, with many caregivers reporting difficulty finding reliable guidance on where to seek services and how to coordinate care. These co-occurring challenges may contribute cumulatively to caregiver stress, time costs, and reduced engagement with services.

Informational barriers were similarly common. Many caregivers reported difficulty finding reliable guidance on where to seek services and how to coordinate care. In our data, information difficulties clustered with lower spouse education and with ADHD and disorders of intellectual development, and caregivers reporting information gaps were more likely to seek guidance from other parents. This

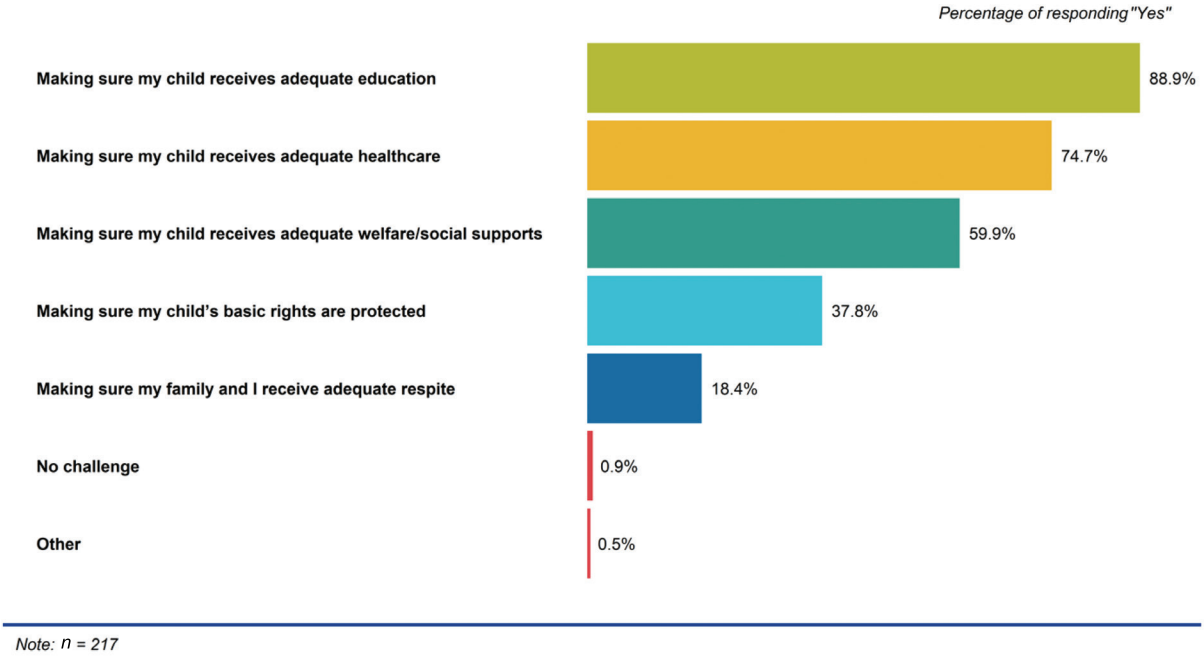

Figure 3: Caregiver-identified challenges in obtaining support for their child. Totals exceed 100% because respondents could select any number of options.

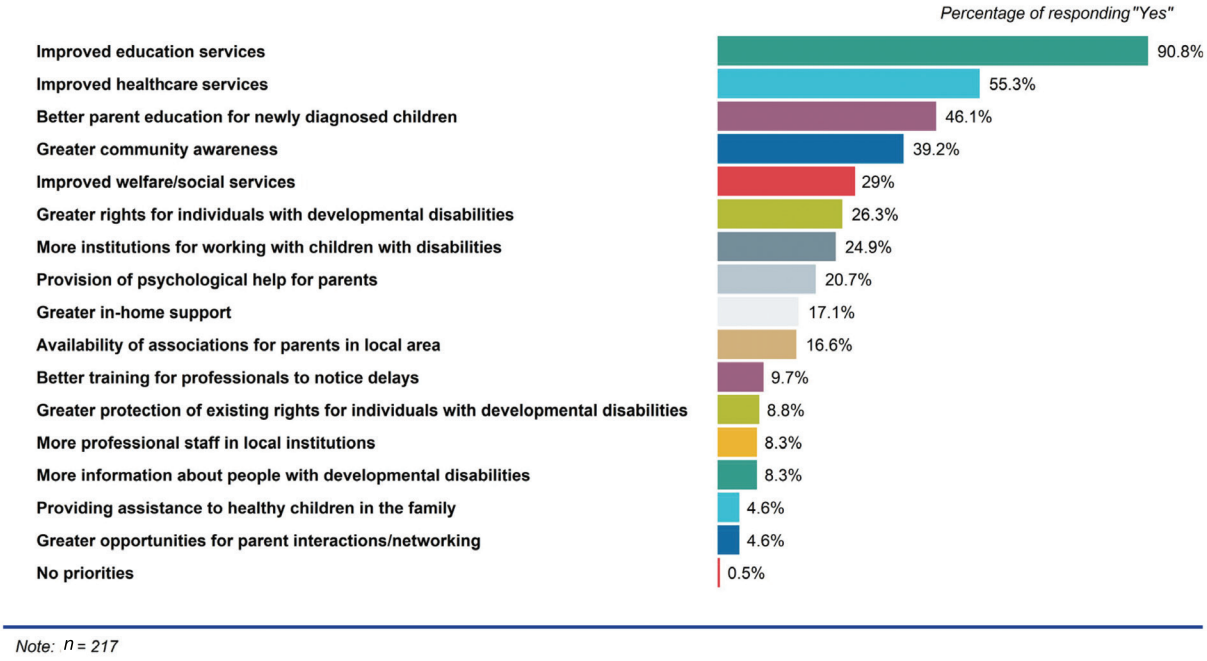

Figure 4: Distribution of caregiver-identified priorities for improving service and support in Jazan. Caregivers of children with developmental conditions indicated their priorities (multiple response). Percentages are based on non-missing responses.

pattern suggests a coping strategy: families relying on peers and informal networks may be attempting to compensate for fragmented official information pathways (Međaković *et al.*, 2024). Saudi caregiver evidence likewise highlights gaps in access to adequate information and coordinated support (Al Awaji *et al.*, 2024). Together, these findings support clearer navigation pathways, standardized counseling at diagnosis, and integrated referral information to reduce both the financial and informational load on families (World Health Organization and UNICEF, 2023).

Caregiver burden and family impact

Caregiver burden was substantial in this sample and closely tied to the service gaps families reported. These levels of

burden likely reflect the cumulative impact of service access barriers, diagnostic delays, and limited support availability. High frustration with obtaining services, financial difficulty, and employment disruption reflect the time, coordination, and opportunity costs of managing a child's NDD-related needs when care is delayed, locally unavailable, or difficult to navigate. Overall, 83.9% of caregivers reported frustration at least sometimes, while 22.6% stopped working due to caregiving responsibilities. These impacts also align with caregivers' stated priorities for improvement, particularly strengthening education services and healthcare access, suggesting that the burden in Jazan is driven by system constraints across healthcare and education and not solely by caregiving demands.

This interpretation is consistent with evidence that unmet needs and fragmented service experiences are linked to poorer family well-being and greater caregiver strain, including in families of children with ASD and co-occurring intellectual disability (Menezes *et al.,* 2021). Saudi studies similarly report elevated psychological distress and reduced quality of life among caregivers of children with NDDs, reinforcing that the burden is both practical and mental health relevant (Alsaad *et al.,* 2023; Alodaili *et al.,* 2026).

System-level guidance emphasizes that reducing family burden requires integrated, family-centered supports across health, education, and social services, underscoring the need for multisectoral interventions (World Health Organization and UNICEF, 2023).

Satisfaction and quality of life

Although access barriers were common, caregivers reported higher satisfaction with professional supports, particularly relationships with providers and practical guidance to support the child's progress at home. By contrast, satisfaction with peer and social support, especially support from friends, was lower. This "support imbalance" is consistent with Saudi caregiver evidence emphasizing the importance of respectful professional care and adequate guidance, while also highlighting how peer connections and support groups may reduce isolation and strengthen caregivers' overall experience (Al Awaji *et al.,* 2024). Similar patterns have been reported elsewhere, where community and informal supports lag behind clinical services and families report unmet needs in day-to-day social support (Međaković *et al.,* 2024). Expanding community-based supports, such as peer groups and structured caregiver networking, could improve caregiver quality of life beyond what clinical service provision alone can offer (World Health Organization and UNICEF, 2023).

Challenges and priorities

Caregivers' priorities in Jazan point to a clear cross-sector service gap across healthcare, education, and social welfare. At the child level, families most often reported behavioral concerns, safety issues, and daily living difficulties; at the family level, the dominant challenges were securing adequate education, healthcare, and welfare or social supports. Provincial priorities echoed these needs, emphasizing strengthening education services first, followed by improved healthcare and therapy access, parent education at diagnosis, and greater community awareness and staff training. These priorities reflect caregiver-identified needs collected in the structured survey (Table 4, Figs. 2-4), highlighting the alignment between reported challenges and desired improvements.

Policy implications

Taken together, these priorities support a multisector response rather than isolated service fixes. In healthcare, the findings argue for expanding locally available NDD services, including therapy capacity, and reducing the access "friction" families experience, such as delays, travel, and unclear pathways. Findings indicating that later diagnosis and complex disorder profiles were associated with multiple access barriers reinforce the need for targeted, multisectoral strategies. Implementation could include service decentralization, enhanced school-linked therapy, structured caregiver navigation support, and cost-reduction initiatives. Saudi caregiver evidence similarly reports that long waiting times and crowded, high-stimulus clinical environments can be stressful for children and families during visits (Al-Jabri *et al.,* 2023).

In education and welfare, caregiver priorities point to the need for practical supports that extend beyond clinic walls. Saudi parents have emphasized accessible schools, academic modifications, and multidisciplinary school-linked services that integrate special education with therapy supports, alongside sustainability barriers that align with this study's cost and capacity themes, including high session costs, shortages of qualified specialists, and limited specialized centers (Al Awaji *et al.,* 2024). Priorities related to awareness and training are also consistent with Saudi evidence calling for comprehensive training to improve autism-related awareness in educational settings and more rigorous training for healthcare providers (Woodman *et al.,* 2025).

Collectively, these findings support coordinated reforms across healthcare, education, and social support to strengthen inclusive service delivery and reduce avoidable caregiver burden, aligned with Saudi Arabia's Vision 2030 priorities for equitable, integrated services.

Strengths and limitations

This study provides among the first systematic evidence on caregiver perspectives in the Jazan region by linking hospital-recorded diagnoses with caregiver reports on service access barriers, unmet needs, and family impact. Using hospital records strengthened diagnostic ascertainment, while the survey captured experiences spanning healthcare, education, and day-to-day family functioning. Several limitations should be noted. First, outcomes were self-reported and may be affected by recall and social desirability bias. Second, the single-center, referral-based sampling limits generalizability

and likely underrepresents children who are not connected to specialty services, including those in rural areas and those managed outside the hospital system. Third, the cross-sectional design and reliance on bivariate analyses preclude causal inference, and observed associations should be interpreted as exploratory. While the study provides descriptive insights, its cross-sectional design and reliance on bivariate analyses preclude causal inference. Findings should be interpreted as exploratory, particularly for subgroup comparisons.

CONCLUSION

Caregivers in the Jazan region reported substantial challenges in accessing healthcare and education for children with NDDs. Service unavailability, prolonged waiting times, financial strain, and difficulties obtaining information were common, with caregiver-reported barriers particularly frequent among families of children with ADHD and disorders of intellectual development. Education emerged as a priority alongside healthcare, as most children received little or no academic support despite complex needs. Caregivers consistently emphasized the need for improvements across health, education, welfare, awareness, and workforce capacity. Addressing these gaps will require coordinated, multisectoral strategies that decentralize services, strengthen inclusive education, and expand family-centered supports to improve equity and access. Future research should explore longitudinal outcomes and assess the impact of targeted policy interventions on caregiver burden and child outcomes.

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CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

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